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UNIVERSITY OF COLOMBO, SRI LANKA



UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Academic Year 2019/2020 – First Year Examination – Semester I – 2020

SCS1207 – Software Engineering I

TWO (2) HOURS

PART A

To be completed by the candidate

Examination Index No:

Important Instructions to candidates:

1. The medium of instruction and question is **English**.
2. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
3. Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
4. Write your index number on each and every page of the Question paper.
5. **This paper consists of two parts, Part A (Question No 1) and Part B (Question No 2, Question No 3 and Question No 4) and submit separately.**
6. Part A has one (1) question and **06** pages. Part B has three (3) questions and **09** pages.
7. Answer **ALL** questions in both Part A and Part B.
8. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
9. **Non-Programmable** calculators are **allowed**.

For Examiner's use only

Question No	Marks
1	
2	
3	
4	
Total	

/

5

Question 1

(a) Define the term Software Engineering.

[2 marks]

Z marks

(b) Classify the following software systems into **systems software** and **application software**.

[5 marks]

Learning management systems	
Compilers	
Operating systems	
Library systems	
Image processing systems	

Index No:

(c) Classify the following software quality attributes into **current usefulness** and **potential usefulness**.

[5 marks]

Reusability	
Portability	
Reliability	
Accuracy	
Security	

(d) What is a software process model?

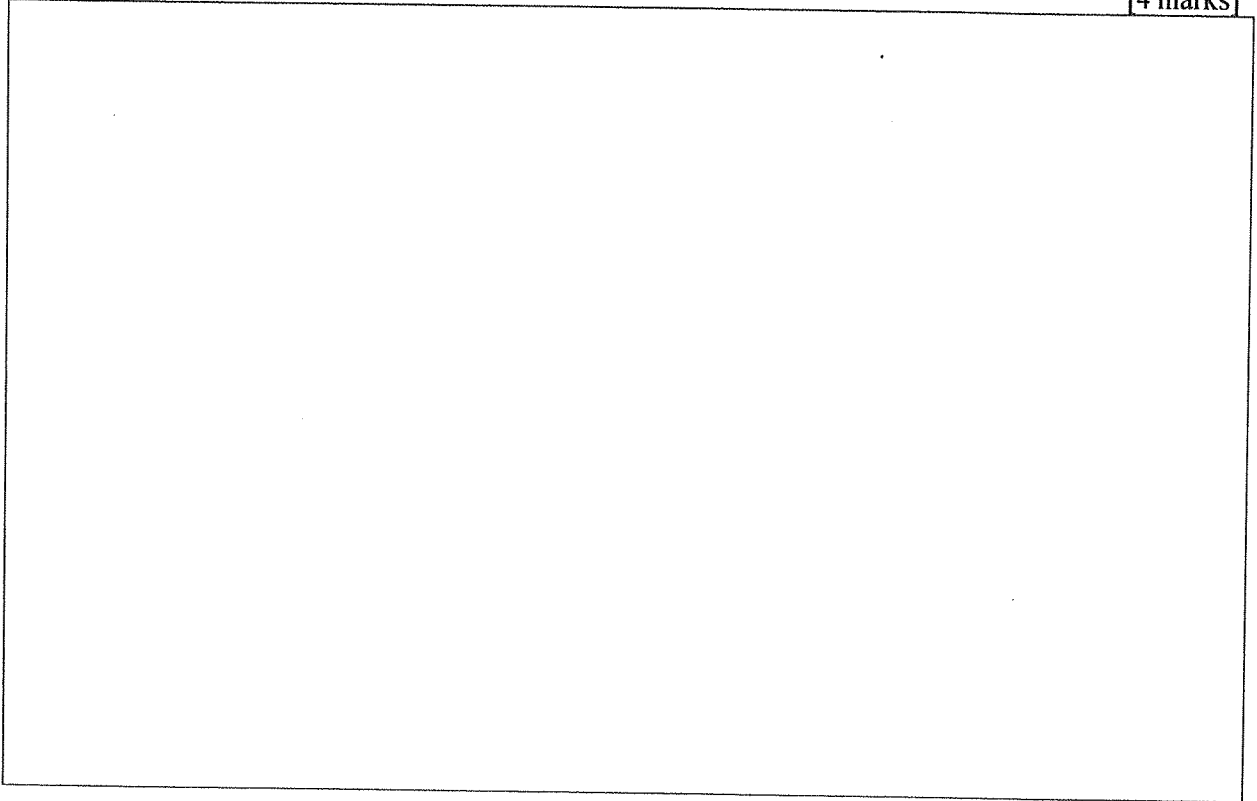
[3 marks]

This image shows a completely blank white page enclosed within a thick black rectangular frame. There are no markings, text, or illustrations on the page itself.

Index No:

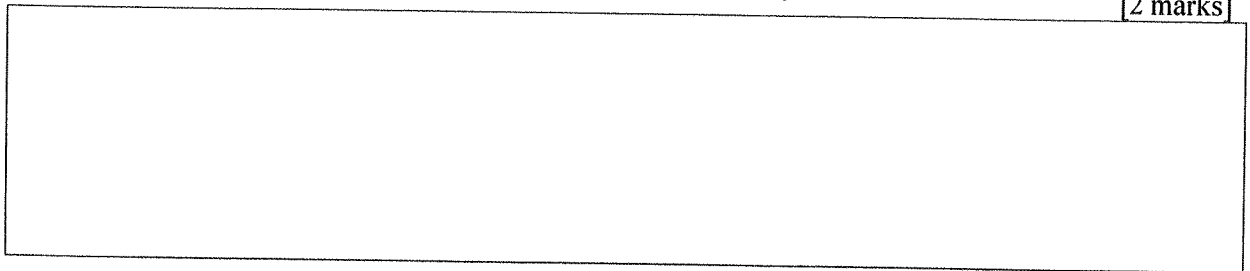
(e) Draw a diagram which indicates the stages of the Waterfall software development model.

[4 marks]



(f) For what type of projects the Waterfall model is suitable?

[2 marks]



(g) Give **three (3)** common features of agile software development methods.

[3 marks]

Index No:

(h) Describe the following two terms with respect to the **Extreme Programming (XP)** method.

[4 marks]

User stories:

Pair programming:

(i) Briefly describe the following three types of requirements in a software development project.

[6 marks]

Functional requirements

Non-functional requirements

Domain requirements

Index No:

- (j) Draw a suitable class relationship diagram for the following classes using UML notation. Indicate the type of relationship.

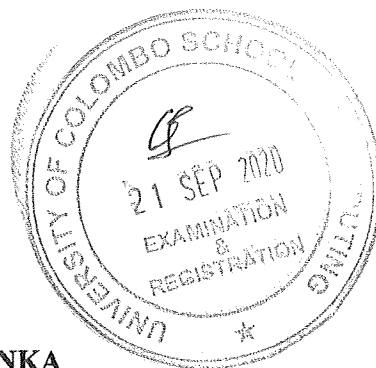
[6 marks]

Vehicle, Car, Van, Lorry

Book, Page

Doctor, Patient

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PART B

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1	
2	
3	
4	
Total	

Index No:

Part B

Question 2

- (a) System modeling is the process of developing abstract models of a software system in different stages of software development.
- (i). Briefly explain the importance of developing models for both the existing system and the system to be developed.

[3 marks]

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- (ii). Briefly explain the below four (4) system modeling perspectives giving each an example of a potential model.

[6 marks]

External Perspective
Interaction Perspective
Structural Perspective
Behavioral Perspective

Index No:

(b) The software architecture is a description of how a software system is organized.

(i). List down **three (3)** advantages of explicitly documenting the software architecture.

[3 marks]

(ii). The choice of architectural style and structure should depend on the non-functional requirements of the system. Justify the statement by providing **two (2)** examples.

[4 marks]

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Index No:

- (c) Briefly explain the difference between an 'Architectural pattern' and a 'Design pattern'. Provide an example for each Architectural pattern and a Design pattern.

[4 marks]

[illegible]

Question 3

- (a) System implementation is a critical stage where an executable version of the system is produced.
- (i). A reuse-based approach is used widely for developing software. List down the **four (4)** levels of software reuse, giving each an example.

[6 marks]

Software Reuse Level	Example

Index No:

- (ii). Identify and write down the configuration management related activities from the below list.
(Host target development, Data reengineering, System integration, System auditing, Release management)

[1 mark]

- (iii). Define the term 'Open Source Development.'

[2 mark]

- (b) Software testing only shows the presence of errors, not their absence.

- (i). Briefly explain in the below two scenarios as to how one can select test cases to define a test is successful.

[3 marks]

Scenario 1: Demonstrate to the customer that the software meets its requirements.

Scenario 2: Discover situations in which the behavior of the software is incorrect.

Index No:

- (ii). Compare and contrast the two testing techniques, 'software inspections & reviews' and 'Defect testing,' by filling the below table.

[5 marks]

	Software Inspections & Reviews	Defect Testing
Similarities		
Differences		

- (c) Briefly explain **three (3)** advantages of test-driven development.

[3 marks]

Index No:

Question 4

- (a) The dosage of a specific drug prescribed by a medical health information system should be between 50 mg – 125 mg. Identify equivalent partitions to test the dosage for the given specification.

[3 marks]

- (b) Mention whether the following statements are TRUE (✓) or FALSE (×). Justify your answer on the space provided.

[6 marks]

- (i). Stress testing is to test the system to make sure that changes have not broken the previously working code. ☐

Justification HERE:

- (ii). Beta-testing is done by the developers using artificial data in the planned operational environment before releasing the system to the client. ☐

Justification HERE:

Index No:

- (iii). The black box testing technique can only be applied at the unit, integration and system levels of the software testing process. ☐

Justification HERE:

Justification HERE:

- (iv). Emergency system repair requests that emerge during the software evolution stage should tackle urgently. ☐

Justification HERE:

Justification HERE:

- (c) Legacy systems are not just software systems but are broad sociotechnical systems.

- (i). List down **four (4)** logical parts of a legacy system on which such systems depend on making the evolution process expensive and risky.

[2 marks]

	Z marks

Index No:

- (ii). Briefly explain the most appropriate strategy for evolving legacy system in the given two situations.

[3 marks]

High-quality system having a low business value

Low-quality system having a high business value

- (d) It is more expensive to add new features to a software system during the maintenance stage. Justify the above statement by providing at least **three (3)** reasons.

[6 marks]
